Midterm Exam

● Graded

Student

Sai Kadiyala

Total Points

20 / 26 pts

Question 1

Q1 **2** / 2 pts

✔ **– 0 pts** Correct answer is C. All other answers would all result in a **less** flexible model.

– 2 pts A, B, D, or E

Question 2

Q2 **2** / 2 pts

✔ **– 0 pts** Correct answer is C. KNN does not make any assumptions about normality or equality of variances, it also doesn't create squared terms.

– 1 pt Partial credit for an answer of B; students could be remembering that this assumption allows QDA to represent nonlinear decision boundaries

– 2 pts A, D, E, or F

Question 3

Q3 **2** / 2 pts

✔ **– 0 pts** Correct answer is B. Moving from Point A to Point C represents an increase in both true positive rate and false positive rate; this corresponds to decreasing the threshold, resulting in more predicted positives overall.

– 1 pt Partial credit for an answer of A (wrong direction, but still recognizing that the threshold would be changing).

– 2 pts C or D.

Question 4

Q4 **1** / 2 pts

– 0 pts Correct answer is Hospital ID. The other variables are specifically defined as unique identifiers, which should not be included as predictors, while patient populations differ across hospital.

✔ **– 1 pt** Partial credit for answering all variables.

– 2 pts A, C, or D

Question 5

Q5 **2** / 2 pts

✔ **– 0 pts** Correct answer is B; if the primary assumption of LDA is being met, fitting QDA will not provide any advantage, and will only increase variance.

– 2 pts A, C, or D

Question 6

Q6 **0** / 2 pts

– **0 pts** Correct answer is B. The definition of an interaction is that the relationship between one variable and the outcome **differs** across levels of the other variable. A change in dosage has more of an effect on Treatment B than it does on Treatment A; this is what suggests an interaction.

✔ – **2 pts** A, C, D, E, or F

Question 7

Q7 **0** / 2 pts

– **0 pts** Correct answer is B. An increase in age of 1 unit will have a positive effect on the log odds (and therefore on the predicted probability), but that will only result in a changing classification if it results in crossing the threshold.

✔ – **2 pts** A, C, or D

Question 8

Q8a **2** / 2 pts

✔ – **0 pts** Sensitivity is the true positive rate (TPR). Increasing the threshold makes it harder to classify observations as positive. This will result in a lower sensitivity.

– **2 pts** Incorrectly saying that sensitivity would increase or stay the same.

– **1 pt** Partial credit

Question 9

Q8b **2** / 2 pts

✔ – **0 pts** Specificity is the true negative rate (or 1 minus the false positive rate). Increasing the threshold will result in a higher specificity.

– **2 pts** Incorrectly saying that specificity would decrease or stay the same.

– **1 pt** Partial credit

Question 10

Q8c **2** / 2 pts

✔ – **0 pts** A change in threshold will have no effect on the ROC curve at all. The ROC curve exists at the model level and is made up of many different thresholds' combinations of true and false positive rates.

– **2 pts** Any other answer.

– **1 pt** Partial credit

Question 11

Q8d **2** / 2 pts

✔ – **0 pts** The number of transactions flagged for investigation will generally decrease as the threshold is raised.

– **2 pts** Any other answer.

Question 12

Q9

3 / 4 pts

- **– 0 pts** Students should recognize that this results in collinearity, regardless of normalization. One predictor will still be a direct function of the other predictor.
- **– 2 pts** Logic is partially correct.
- **– 4 pts** Logic is completely incorrect.

✔ **– 1 pt** Correct answer (that the reasoning is wrong) but doesn't specifically mention collinearity

PSTAT 131 Midterm Exam

Summer 2026

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First and Last; Please Print Clearly!

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This is what comes before the @ucsb.edu in your UCSB Email Address

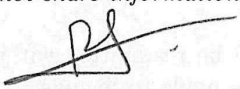
Your TA (circle one): **Chunyin** **Mallory**

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1. **Do not open this exam until you are told. Do not remove this cover sheet.**
2. **Identification:** Fill in your full name and UCSB NetID in the spaces above
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7. **Timekeeping:** If a smart watch is used for timekeeping, it must be in airplane mode with all wireless communication (Wi-Fi/Bluetooth) disabled.
8. **Final Answers:** Round answers to 3–4 decimal places when appropriate. Answers must be legible; if multiple answers are given, the first will be graded, except for multiple choice questions; unless otherwise specified, multiple answers marked for Multiple Choice questions will receive a score of zero.
9. **Restroom Breaks:** If you must leave the room, give your exam paper and phone to the proctor to hold onto while you are out.
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11. **Submission:** Turn in your exam to a proctor upon completion or when time is called. Show photo ID.

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In signing my name below, I certify that all work on this exam is my own. I affirm that I have not used any unauthorized aids, technology (including generative AI tools), or electronic devices, nor have I done anything to create an unfair advantage for myself or any other course member. I have not shared and will not share information with any other student(s).



Student Signature

Multiple Choice Questions

1. (2 points) Which change generally makes a statistical learning method **more flexible**?

   (A) Increasing k in KNN

   (B) Removing predictors from a linear model

   (C) Adding polynomial or interaction terms to a regression model

   (D) Using LDA instead of QDA

   (E) All of the above

2. (2 points) Why can KNN represent nonlinear decision boundaries even without explicitly including polynomial or interaction terms?

   (A) It assumes predictors are normally distributed

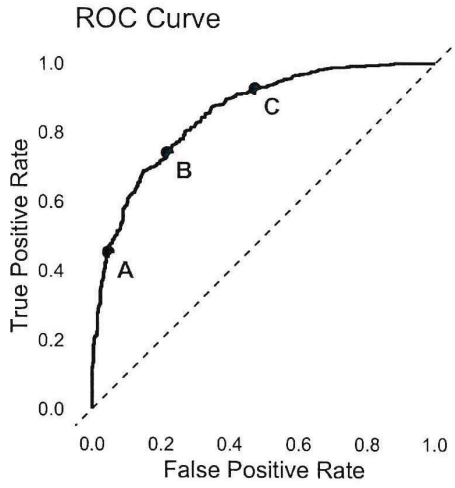
   (B) It does not assume equal variances across levels of the outcome

   (C) Classification is determined locally by neighboring observations

   (D) It estimates a different regression coefficient for every observation

   (E) It automatically creates squared terms

   (F) Both A and B



3. (2 points) Moving along the ROC curve from Point A toward Point C corresponds generally to:

   (A) Increasing the classification threshold

   (B) Decreasing the classification threshold

   (C) Increasing the AUC

   (D) Decreasing the AUC

4. (2 points) A hospital is developing a model to predict whether a patient will be readmitted within 30 days after discharge. Which of the following variables would be most reasonable to include as a predictor?

(A) `PatientID`, a randomly assigned unique identifier for each patient.

(B) `HospitalID`, which identifies which of 15 hospitals treated the patient. Hospitals differ in their patient populations.

(C) `AdmissionID`, a unique identifier assigned to each hospital admission.

(D) `RecordID`, a sequential identifier (1, 2, 3, ...) assigned to each row of the dataset.

(E) None of the above.

5. (2 points) Suppose the population covariance matrices really are identical across all classes. What advantage does QDA gain from estimating a separate covariance matrix for every class?

(A) A closer fit to the population decision boundary

(B) No inherent advantage from the extra covariance flexibility; it may simply increase estimation variance

(C) Perfect classification

(D) QDA, unlike LDA, is nonparametric

6. (2 points) Consider the following fitted values:

	Treatment A	**Treatment B**
Low dosage	20	25
High dosage	30	50

Which feature of these results most strongly suggests an interaction?

(A) Treatment B always has the larger fitted value

(B) Increasing dosage changes the fitted value by 10 under Treatment A but by 25 under Treatment B

(C) All four fitted values are different

(D) Increasing dosage increases the fitted value across both treatments

(E) Dosage has two levels

(F) Both B and D

7. (2 points) In a logistic regression model, the coefficient for `age` is positive. Increasing one customer's age from 40 to 41 does not change that customer's predicted class. Why is this possible?

(A) A positive coefficient does not affect predicted probabilities

(B) The predicted probability can increase without crossing the classification threshold

(C) Logistic regression coefficients affect only training observations

(D) Age must interact with another predictor before it can affect classification

Free Response Questions

8. A bank uses a model to flag credit-card transactions as potentially fraudulent. The classification threshold previously used was 0.40; the bank's lead data scientist decides to increase the threshold to 0.70.

   Explain the effect, if any, that this change would generally have on each of the following:

   (a) (2 points) Sensitivity

   Sensitivity would be lowered

   as high sensitivity → crease a point with high TPR even if FPR is higher → this means lower threshold

   so

   we are doing the opposite here

   (b) (2 points) Specificity

   Being stricter would decrease FPR and the formula for specificity is $1 - FPR$ so if FPR gets smaller specificity would go up.

   (c) (2 points) The model's ROC curve

   The model's curve wouldn't necessarily change, only the dot of the threshold would. Only determines which point on the ROC curve you operate at.

   (d) (2 points) The overall number of transactions flagged for investigation

   when the threshold is increased, it means we require more "certainty" (higher predicted probability) - so fewer observations exceed the high bar, so we get fewer positive predictions, more negative predictions.

9. (4 points) Consider the following recipe. The model's goal is to predict the sale price of a home. `square_feet` represents the house size measured in feet; `square_meters` represents the house size measured in meters.

```
house_recipe <-
  recipe(sale_price ~ ., data = houses) %>%
  step_mutate(
    square_meters = square_feet * 0.092903
```

```
) %>%
step_normalize(all_numeric_predictors())
```

A student argues that because both `square_feet` and `square_meters` are normalized, it is appropriate to keep both in an ordinary linear regression. Is this reasoning correct or incorrect? Explain.

In simple linear regression we have this formula: $Y = \beta_0 + \beta_1 X + \epsilon$ ← error/noise

- Y ↓ response variable
- β_0 ↘ intercept
- β_1 ↗ slope

The point I'm getting to is that, I personally believe it is not appropriate because the resulting value would be the same (if only one of the two is used). If you normalize both in terms of price your result would be same as standardizing/normalizing means $\mu = 0$ $\sigma = 1$ in a ∩ normaly distributed curve. so I believe it isn't.